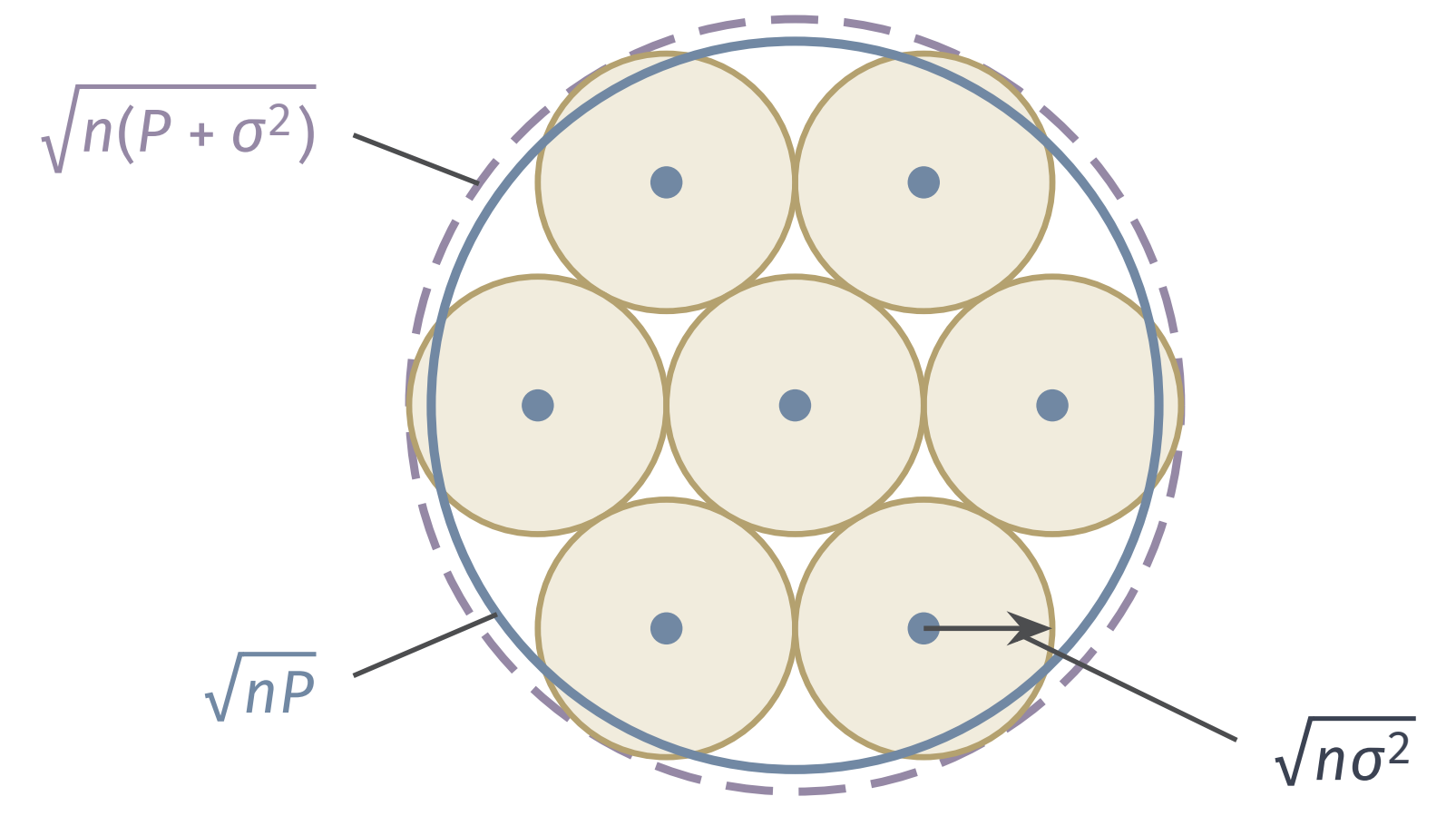
$\sqrt{n(P+\sigma^2)}$
$\sqrt{nP}$
$\sqrt{n\sigma^2}$